



03 A First Problem: Stable Matching

CS216 Algorithm Design and Analysis (H)

Instructor: Shan Chen

chens3@sustech.edu.cn



Stable Matching

- **Motivation:** [Gale and Shapley 1962]
 - Can one design a college admissions process, or job recruiting process, that is **self-enforcing (stable)**?
- **Example:** A “bare-bones” version of job recruiting
 - m applicants and n companies
 - Each applicant ranks companies and each company ranks applicants
 - Each company may hire **one or multiple** applicants.
- Let's first look at a **simpler** setting: **one-to-one matching** (e.g., **marriage**)
 - n men: $A = \{m_1, m_2, \dots, m_n\}$ and n women: $B = \{w_1, w_2, \dots, w_n\}$
 - Each man can be married to **at most one** woman and vice versa.
 - A matching M is a subset of the Cartesian product $A \times B$.



Some Definitions

- **Perfect Matching:** everyone is matched **monogamously** (一夫一妻)
 - Each man **gets exactly one** woman.
 - Each woman **gets exactly one** man.
- **Stability:** **no pair** of participants has incentive to **undermine the current matching by joint action**
 - In a matching M , an **unmatched** pair $m - w$ is **unstable** if man m and woman w prefer each other to their current partners.
 - An unstable pair $m - w$ could each improve by joint action (e.g., eloping).
- **Stable Matching:** **perfect** matching with **no unstable pairs**



The Stable Marriage/Matching Problem

- **The Stable Marriage/Matching Problem.** Given the preference lists of n men and n women, find a stable matching if one exists.
- **Example 1 ($n = 2$):** [$m_1: w_1 > w_2; m_2: w_1 > w_2; w_1: m_1 > m_2; w_2: m_1 > m_2$]
 - The stable matching $\{m_1 - w_1, m_2 - w_2\}$ is unique.
 - ✓ The other perfect matching $\{m_1 - w_2, m_2 - w_1\}$ has an unstable pair $m_1 - w_1$.
- **Example 2 ($n = 2$):** [$m_1: w_1 > w_2; m_2: w_2 > w_1; w_1: m_2 > m_1; w_2: m_1 > m_2$]
 - Both perfect matchings are stable:
 - ✓ $\{m_1 - w_1, m_2 - w_2\}$: both men are happy
 - ✓ $\{m_1 - w_2, m_2 - w_1\}$: both women are happy



Questions

- **Q.** Do stable matchings always exist in general?
- **A.** No. See the counterexample (not a marriage problem) below.

	1 st	2 nd	3 rd
A	B	C	D
B	C	A	D
C	A	B	D
D	A	B	C

no perfect matching is stable:

$\{A - B, C - D\} \Rightarrow B - C$ unstable

$\{A - C, B - D\} \Rightarrow A - B$ unstable

$\{A - D, B - C\} \Rightarrow A - C$ unstable

- **Q.** Does there exist stable matchings for the marriage problem?
- **Q.** How can we find such a stable matching?



The Gale-Shapley Algorithm

- **The Gale-Shapley algorithm.** [Gale-Shapley 1962] An intuitive method that **guarantees to find a stable matching**.
 - also known as the **propose-and-reject** or **delayed-acceptance** algorithm
 - Idea: **men propose** to preferred women (but may get rejected) until all matched

```
Initialize each person to be free.
while (some man is free and hasn't proposed to every woman) {
    Choose such a man m
    w = 1st woman on m's list to whom m has not yet proposed
    if (w is free)
        assign m and w to be engaged
    else if (w prefers m to her fiancé m')
        assign m and w to be engaged, and m' to be free
    else
        w rejects m
}
```



Proof of Correctness: Termination

- **Observation 1.** Men propose to women in **decreasing preference order**.
- **Observation 2.** Women only **“trade up”**: once a woman is matched, she never becomes unmatched.
- **Claim.** Algorithm terminates after at most n^2 iterations of the while loop.
- **Pf.** Each time through the while loop a man proposes to a new woman. There are only n^2 possible proposals. ■
- **Q.** Can you think of a scenario that requires $\Theta(n^2)$ steps for GS?
- **A.** E.g., all men ranks women in the same order, and all women ranks men in the opposite order.



Proof of Correctness: Perfection

- **Claim.** All men and women are uniquely matched.
- **Pf. (by contradiction)**
 - Suppose that some man Zeus is **not matched** upon termination of algorithm.
 - Then some woman, say Amy, is not matched upon termination.
 - By **Observation 2**, Amy **was never proposed to**.
 - But Zeus **proposed to everyone**, since he ends up unmatched. Contradiction! ■



Proof of Correctness: Stability

- **Claim.** There exist no unstable pairs.
- **Pf. (by contradiction)**
 - Suppose $Z - A$ is an **unstable** pair (see the bottom-right figure), i.e., each prefers each other to their current partner in the Gale-Shapley matching S^* .
 - Case 1: Z never proposed to A .
 - ✓ Z prefers B to A . ← men propose in decreasing order of preference
 - ✓ So, $Z - A$ is stable.
 - Case 2: Z proposed to A but got rejected (right away or later)
 - ✓ A prefers Y to Z . ← women only trade up
 - ✓ So, $Z - A$ is stable.
 - In either case, $Z - A$ is **stable**. Contradiction! ■

S^*

Zeus - Bertha
Yancey - Amy
...



Quick Summary and Questions

- **The Stable Marriage Problem.** Given n men and n women, and their preferences, find a stable matching if one exists.
- **The Gale-Shapley algorithm.** Guarantees to find a stable matching for **any** problem instance.
- **Q.** How to implement the GS algorithm efficiently?
- **Q.** If there are multiple stable matchings, which one does GS find?



Efficient $O(n^2)$ Implementation

- **Representing men and women.**

- Assume men and women are each named $1, \dots, n$.

- **Recording the matching.**

- Maintain a list of free men, e.g., in a **queue or stack**.
- Maintain two arrays **wife[m]**, and **husband[w]**.
 - ✓ Set entries to **0** if unmatched.
 - ✓ If m matches w , then **wife[m] = w** and **husband[w] = m**.

- **Men proposing.**

- For each man, maintain a list of women, ordered by preference.
- Keep an array **count[m]** that counts the number of proposals made by man m .



Efficient $O(n^2)$ Implementation

- **Women accepting/rejecting.**

- How can we **efficiently** check if woman w prefers man m to man m' ?
- For each woman, create an **inverse mapping** from men to preference orders.
✓ $O(1)$ access for each query after $O(n)$ preprocessing

Amy	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th
pref	8	3	7	1	4	5	6	2

```
for i = 1 to n
    inverse[pref[i]] = i
```

Amy	1	2	3	4	5	6	7	8
inverse	4 th	8 th	2 nd	5 th	6 th	7 th	3 rd	1 st

E.g., Amy prefers man 3 to 6 since
 $\text{inverse}[3] = 2 < 7 = \text{inverse}[6]$

- **Memory.** It is not hard to see that GS consumes $O(n^2)$ memory.

- Input and output also take memory!



Understanding the Solution

- **Q.** For a given problem instance, there may be several stable matchings. Do all GS executions yield the **same** stable matching? If so, **which one**?
- **Def.** Man m is a **valid partner** of woman w if **there exists** some **stable** matching in which they are matched.
- **Def.** The **man-optimal** matching: **every** man receives **best** valid partner.
- **Claim.** All GS executions yield the **man-optimal** matching, which is also a **stable** matching! Very surprising, isn't it?
 - The man-optimal matching is **simultaneously best** for all men.
 - No reason to believe that the man-optimal matching exists, let alone stable!



Man Optimality

- **Claim.** The GS matching S^* is **man-optimal**.
- **Pf.** (by contradiction)
 - Suppose S^* is not man-optimal, i.e., some man is not paired with his best valid partner. Since men proposed in decreasing order of preference, **some man is rejected by his valid partner**.
 - Let Y be the **first** such man and let A be the **first valid partner** of Y that rejects Y .
 - When Y is rejected, A (re)affirms matching with a man, say Z , whom A prefers to Y . We know Z was not rejected by any valid partner at this point, so Z prefers A to any other valid partners. ↙ since A is a valid partner of Y
 - There exists a **stable** matching S where Y and A are matched. Let B be Z 's valid partner in S . From above, Z prefers A to B .
 - Also, A prefers Z to Y , so $Z - A$ is **unstable** in S . Contradiction! ■

S

Yancey-Amy

Zeus-Bertha

...



Woman Pessimality

- **Q.** Does man-optimality come at the expense of the women?
- **Def.** **Woman-pessimal** assignment: **every** woman gets **worst** valid partner.
- **Claim.** The GS matching S^* is **woman-pessimal**.
- **Pf.** (by contradiction)
 - Suppose $Z - A$ is matched in S^* , but Z is not the worst valid partner for A .
 - There exists a **stable** matching S in which A is paired with a man, say Y , whom A likes less than Z . Let B be Z 's valid partner in S .
 - From **man-optimality** of S^* , we have Z prefers A to B .
 - Recall A prefers Z to Y , so $Z - A$ is **unstable** in S . Contradiction! ■

S	
Yancey-Amy	
Zeus-Bertha	
...	



Gale-Shapley Algorithm: Summary

- **The Gale-Shapley algorithm.** Finds a stable matching in $O(n^2)$ time.
- **Man Optimality.** In the version of the GS algorithm where men propose, each **man** receives **best valid partner**.
- **Woman Pessimality.** In the version of the GS algorithm where men propose, each **woman** receives **worst valid partner**.
- **Q.** If you want a best mate, would you propose or wait to be proposed?



Extension: Matching Students to Hospitals

- **Extension:** hospitals hire medical students
 - Variant 1. Participants declare others as **unacceptable**. ← some student is unwilling to work in some hospitals, or the other way.
 - Variant 2. **Unequal** number of positions and students.
 - Variant 3. Limited **polygamy**.
← some hospital could hire multiple students, e.g., ≤ 3
- In **Assignment 1**, you will be asked to prove that Gale-Shapley can be adapted to find stable matchings in the above generalized setting.
 - To prove it, you need to first define **stable matching** in this setting.



Men/Women $\neq$ Hospitals/Students

- **Men/Women Marriage:** one-to-one matching
- **Hospitals/Students Recruitment:** one-to-many matching
- For around 20 years, most people thought the above problems had very similar properties. However, this is wrong.
 - **[Roth 1982]** Any algorithm for men/women marriage (e.g., man-proposing GS) that yields a man-optimal stable matching implies that truth telling is the dominant strategy for men.
 - **[Roth 1985]** No stable matching algorithm for hospitals/students recruitment exists such that truth-telling is the dominant strategy for hospitals.





Real-World Application: NRMP

- **National Resident Matching Program (NRMP):**

- The algorithm is an extension to GS but was in practical use before GS!
- Original use in 1950s, just after WWII. ← **predates computer usage**
- Initial version does not handle couples and other special cases.
- The full algorithm was adopted and used since late 1990s.

- **Rural hospital dilemma.** Certain hospitals (mainly in rural areas) are unpopular and declared unacceptable by many students.

- How can we find stable matchings that benefit “rural hospitals”?

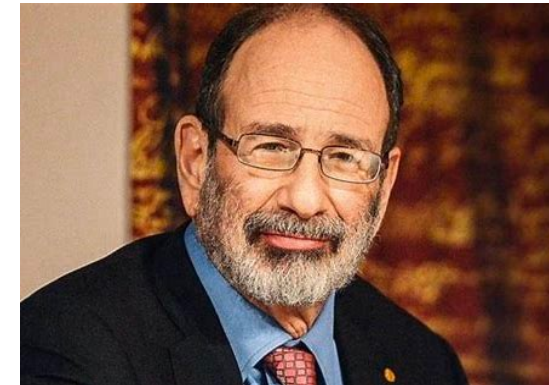
- **Rural hospital theorem.** [Roth 1986] Rural hospitals get **exactly the same** students in every stable matching!





2012 Nobel Prize in Economics

- **Lloyd Shapley.** Stable matching theory and Gale-Shapley algorithm.
- **Alvin Roth.** Applied Gale-Shapley to matching med-school students with hospitals, students with schools, and organ donors with patients.





More on Stable Matching

- **The stable roommate problem:**

- Matching is defined on **general** graphs (may be **non-bipartite**)
- Stable matchings may not exist!

- **Q.** Can we find a **polynomial-time** algorithm that does the following?

- either finds a stable matching
- or **reports non-existence**

- **A.** Irving's algorithm [**Irving 1985**]

- builds on GS ideas and work by [**McVitie and Wilson 1971**].





Lessons Learned

- **Powerful ideas of algorithm design and analysis:**
 - Isolate underlying structure of the problem.
 - Design useful and efficient algorithms.
 - Prove correctness and bound time and memory.
- **Caveat.** Potentially deep **social ramifications**. [legal disclaimer]



Announcements

- **Assignment 1 has been released and the deadline is Mar 18.**
- **Lab 2 will be released today and the deadline is Mar 18.**